

Bachelor in Computer Applications

Course Title: **Artificial Intelligence**

Code No: Semester: **4**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. List the properties that a machine must have to pass the Turing Test. Describe the ambiguities in each step in Natural Language Processing.
2. Define agent. How do you describe PEAS for an agent? Describe the different types of environment with example.
3. How do Depth Limited Search solve the problem of Depth First Search and discuss about its limitations? Illustrate the concept of AO* search in multiplication of any 3 matrices.
4. Explain the architecture of expert system.
5. How does Genetic algorithm work? Explain.
6. Discuss different types of Neural Network.
7. What is machine learning? Describe about NLU and NLG.
8. Distinguish between backward chaining and forward chaining with an example.
9. Define game. Give a state space representation of any problem you know.
10. Represent the following sentences into semantic network.
a. All animals and plants are living things.
b. Rose is a plant.
c. All plant prepare food using photosynthesis process.
d. Carnivorous animal don't eat plant.
e. Tiger is carnivorous.
11. Explain about belief network with an example.
12. Discuss about Hill climbing search with its limitations.